CSA4707 – Deep Learning

Experiment 1: Confusion Matrix

Aim:

To demonstrate a confusion matrix using Python.

Program:

```
import numpy as np
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

actual = np.array(['Dog', 'Dog', 'Dog', 'Not Dog', 'Dog',
                   'Not Dog', 'Dog', 'Dog', 'Not Dog', 'Not Dog'])

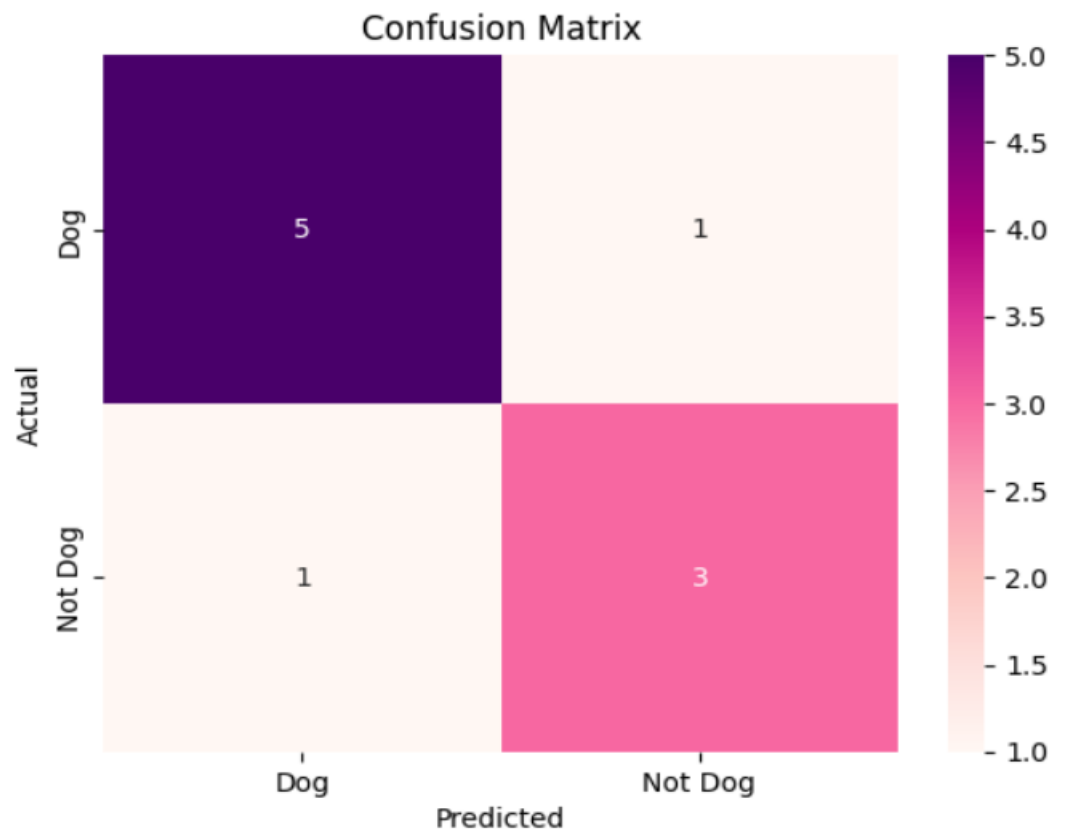
predicted = np.array(['Dog', 'Not Dog', 'Dog', 'Not Dog', 'Dog',
                      'Dog', 'Dog', 'Dog', 'Not Dog', 'Not Dog'])

conf_matrix = confusion_matrix(actual, predicted, labels=['Dog', 'Not Dog'])

sns.heatmap(conf_matrix, annot=True, fmt='g',
            xticklabels=['Dog', 'Not Dog'],
            yticklabels=['Dog', 'Not Dog'], cmap='RdPu')

plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.title("Confusion Matrix")
plt.show()
```

output:



Result:

The confusion matrix was successfully generated using Python to compare actual and predicted classifications.